

Markscheme

Mathematics: analysis and approaches

Practice question bank

56. (a)

A radioactive sample decays according to $M = M_0(0.5)^{\frac{t}{5730}}$, where t is measured in years. A sample originally of mass 80 g has decayed to 15 g.

Find the age t of the sample, correct to the nearest 10 years.

set up the equation: $\frac{15}{80} = (0.5)^{\frac{t}{5730}}$ (M1)

take $\ln$ of both sides: $\ln(0.1875) = \frac{t}{5730} \ln(0.5)$ (M1)

$$t = 5730 \times \frac{\ln(0.1875)}{\ln(0.5)} \quad \text{A1}$$

$$t = 13838.4 \dots \quad \text{A1}$$

$$t \approx 13840 \text{ years} \quad \text{A1}$$

[5 marks]